Boshi Wang

+86 13691174556 | wbsstonewang@126.com | Shanghai, China

Education

ShanghaiTech University Shanghai, China
B.E. in Computer Science and Technology Sep. 2016 - Present

- **GPA**: 3.76/4.0 (Rank 5/116)
- **Graduate-level Coursework**: Convex Optimization, Matrix Analysis, Algorithmic Game Theory, Subspace Learning, Topological Data Analysis
- **Programming Skills**: C/C++, Python, Matlab, Rust, MIPS assembly
- **Tools**: CVX, Robot Operating System (ROS), Spark, Tensorflow, Pytorch

Publications/Papers

- **B. Wang**, Y. Guo, D. Hughes, K. Sycara. Developing Ethical Agents via Context-Sensitive Modular Inverse Reinforcement Learning. In *Carnegie Mellon Robotics Institute Summer Scholars (RISS) Working Papers Journal*, 2019.

- **B. Wang**. Homomorphic Sensing in Union of Linear Subspaces with Applications in Sparse Signal Recoveries. (In Preparation**,** ***Link to Paper***)

- Z. Tu, **B. Wang**, T. Duan, and F. Gao. Pattern-learning-based Noise Elimination Algorithm in Photoacoustic Sensing and Imaging. In *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'18)*, Honolulu, HI, USA, on July 17-21, 2018.

Research Experience

Advanced Agent-Robotics Technology Lab, Carnegie Mellon University Pittsburgh, USA
Undergraduate Research Intern, supervised by Prof. *Katia Sycara* Jun 2019 - Sep 2019

- Developed Context-Sensitive Modular Inverse Reinforcement Learning (CMIRL), a computational scalable Inverse-RL framework for developing norm-compliant autonomous agents through learning from demonstrations under large number of environmental contexts. Link to *Poster*.

Manolis Research Group, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Manolis C. Tsakiris* Feb 2019 - Present

- Extended Homomorphic Sensing (a recent ML research topic proposed by Prof. Tsakiris in an *ICML'19* paper) into the case where the input space is a union of finitely many linear subspaces instead of a single linear subspace, which gives conditions for uniqueness guarantees for a class of sparse signal recovery problems.

- Currently conducting theoretical analysis on subspace recovery under partially permuted data with Dual Principal Component Pursuit (a non-convex non-smooth Robust Subspace Learning method), and also on probabilistic analysis of noisy Homomorphic Sensing.

Hybrid Imaging System Laboratory, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Fei Gao* Jan 2018 - Nov 2018

- Developed an algorithm that can learn from historical data about the pattern of a periodic noise in Photoacoustic signals and efficiently perform noise reduction.
- Experimental study on using denoising autoencoders (DAE) for photoacoustic signal noise reduction.

Mobile Autonomous Robotic Systems Lab, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Sören Schwertfeger* Jun 2017 - Dec 2017

- Implemented Visual Odometry using Aruco markers in the surrounding environment, and combined it with motion-based odometry (developed by collaborators) via G2O to achieve high localization accuracy. The work contributes to a bigger project in the lab about Simultaneous Localization and Mapping (SLAM).

Teaching Experience

Teaching Assistant of *SI140: Probability and Mathematical Statistics*, ShanghaiTech University.
Course Instructor: Prof. *Ziyu Shao* Shanghai, China

Industry Experience

Chinese Software Developer Network Corporation (CSDN) Beijing, China
Intern of System Developer Jun 2018 - Sep 2018

- Optimized the keyword matching and filtering algorithm of the blog recommendation system.
- Developed an efficient method for clustering search queries of blogs and implemented the method with parallel computing using Spark.

Other Selected Projects

Deep Q-learning based AI agent for Beating the Landlord
Course project of *CS181: Artificial Intelligence* in ShanghaiTech University, Fall 2018

- Developed a Deep Q-learning framework which learns to play a popular 3-player card game in China called "Beating the Landlord" through three agents' self-play. The trained agent largely outperforms greedy agents which are used in most online game platforms.

Sparse Representation Based Face Recognition
Course project of *SI131: Linear algebra* in ShanghaiTech University, Fall 2017

- Implemented a face recognition algorithm based on Principle Component Analysis and Sparse Representation based classification. The algorithm achieves 91.4% accuracy on Extended Yale Face Database B.